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A SIEM is the single core operating system of a modern SOC. It is a massive data lake designed for real-time streaming analytics, long-term security retention, and forensic compliance.

In a Security Operations Center (SOC), the **SIEM (Security Information and Event Management)** is the brain. If an EDR gives you sight into endpoints, the SIEM gives you sight into the *entire enterprise*—merging logs from networks, clouds, firewalls, and applications into one single pane of glass



SOC OPERATIONAL SIEM DASHBOARD: ENTERPRISE VIEW

THREAT LANDSCAPE OVERVIEW

GLOBAL ALERT SEVERITY



TOP ATTACK VECTORS



ALERT VOLUME (LAST 24 HOURS)



INCIDENT QUEUE & INVESTIGATION

PRIORITY INCIDENTS

ID	SEVERITY	USER/HOST	STATUS	ASSIGNED TO
17701	CRITICAL	SQL_PROD	Triage	Bark Sadtson
17702	HIGH	USER/HOST	In Progress	Elf Finance
17733	HIGH	WEBSERVER_02	Contained	Admim Sherson
17704	CRITICAL	USER/HOST	Contained	Aewkwnrth

MEAN TIME TO ACKNOWLEDGE (MTTA)



MEAN TIME TO RESPOND (MTTR)



DATA SOURCES & INGESTION HEALTH

LOG INGESTION PIPELINE



INGESTION RATE BY SOURCE (GB/DAY)



EPS (EVENTS PER SECOND)



LICENSE UTILIZATION



ENTITY BEHAVIOR & HUNTING

GEO-IP ANOMALIES



UNUSUAL LOGINS

TOP AT-RISK ENTITIES

USERS	SCORE	HOSTS	SCORE
VP Finance	High risk	SQL_PROD	High risk
Dev Lead	High risk	WEBSERVER_02	High-risk
Dev Lead	High risk	WEBSERVER_02	High risk

THREAT HUNTING SEARCH

Search...

Suggested queries: SEARCH
Suggested queries: SQL_PROD_PRODS
Suggested queries: SQLPROD
Suggested queries: WEBSERVER_02

The Architectural Blueprint (The Ingestion & Processing Pipeline)



A SIEM is essentially a giant machine that ingests unstructured or semi-structured data, normalizes it on the fly, and optimizes it for ultra-fast text search. Splunk achieves this via different distinct architectural stages:

Stage A: Collection (The Edge Forwarders)

Logs originate at the edge of the network. Splunk uses **Universal Forwarders (UFs)**—highly optimized, C++ binaries that run with almost zero CPU overhead on endpoints, servers, and domain controllers.

- **The Mechanism:** The UF uses configuration templates (primarily `inputs.conf`) to monitor specific directories or event channels (e.g., `WinEventLog://Security`). It reads the log line, wraps it in a lightweight framing protocol, and stream-shunts it over TCP port **9997**.

The Architectural Blueprint (The Ingestion & Processing Pipeline)



Stage B: Heavy Parsing & Data Reduction (The Traffic Cop)

Before data hits long-term disk storage, it passes through a **Heavy Forwarder (HF)** or intermediate tier. This is a critical stage for a SOC's financial and performance budgeting.

- **Data Filtering:** A SOC engineer will write Regex rules inside `props.conf` and `transforms.conf` to discard millions of junk logs (e.g., routine firewall "allowed" connection logs or debug noise) before they touch the indexer. This saves millions of dollars in ingestion licensing costs.
- **Anonymization & Masking:** If a log contains sensitive personal information (PII) like cleartext social security numbers or credit cards, the HF intercepts it and uses Regex to mask it (XXXX-XX-1234) to comply with privacy frameworks (GDPR/PCI-DSS).



Stage C: Indexing & Storage (The Time-Series Database)

The **Splunk Indexer** transforms raw text streams into highly structured, searchable data. This is where the heavy computational lifting happens:

- **The Parsing Phase:** The indexer breaks the text stream into discrete events, extracts the explicit UTC timestamp (`_time`), and tags the data with core metadata: `host` (where it came from), `source` (the file pathway), and `sourcetype` (the structural format, like `cisco:asa` or `xmlwineventlog`).
- **The Bucket Lifecycle:** Data is written to time-series directories on disk called **Buckets**.
 1. **Hot Buckets:** Actively being written to; held in super-fast storage (NVMe/SSD). Searchable.
 2. **Warm Buckets:** Full hot buckets rolled over. Still on fast storage, searchable, but no longer receiving new entries.
 3. **Cold Buckets:** Older data moved to cheap, high-capacity mechanical or cloud object storage. Searching takes significantly longer.
 4. **Frozen Buckets:** Expired data. Deleted entirely or compressed and archived off-site (unsearchable without manual re-indexing).

Data Normalization: Splunk CIM (Common Information Model)



Raw security logs from different tools use completely different naming conventions for the exact same concepts.

If an enterprise uses **Palo Alto Firewalls**, **Zscaler Web Proxies**, and **AWS VPC Flow Logs**, a routine connection block looks entirely different across all three systems. If an analyst wants to write a rule to check for a blocked IP across the entire company, they would have to write a massively complex query accounting for every single vendor's specific vocabulary.

Palo Alto Log —> `action=deny src=192.168.1.50`

Zscaler Log —> `block_type=policy_blocked client_ip=192.168.1.50`

AWS VPC Log —> `log_status=REJECT srcaddr=192.168.1.50`



The Solution: Field Aliasing & Tagging

Splunk solves this via the **Common Information Model (CIM)**. During search-time execution, data-model mappings run in the background. Splunk converts all those messy vendor variables into standard, universal fields:

```
Action = "blocked"  
src_ip = "192.168.1.50"
```

Because the data is normalized, a SOC analyst can write one single, highly concise query that filters across all security vendors simultaneously:

```
tag=network tag=communicate action=blocked src_ip=192.168.1.50
```


The Alerting Process: Real-Time Correlation Engines



Once logs are securely indexed and normalized, the SIEM switches from a passive repository to an active, real-time threat detection engine. This is handled by **Correlation Searches** running inside **Splunk Enterprise Security (ES)**.

These searches execute on an automated background cron-schedule (e.g., every 5 minutes, checking the previous 5 minutes of logs) using **SPL (Search Processing Language)**.

The Incident Review & Analysis Lifecycle

When that correlation search meets its specific parameters, it fires a **Notable Event** inside **Splunk Enterprise Security's** triage pane. This kicks off the core human operations pipeline of the SOC.

The **SIEM** automatically parses out the core fields involved (**user, src_ip, host**) and cross-references them against internal corporate **Lookups** (Watchlists) in real-time. Before an analyst even clicks on the ticket, the **SIEM** has already flagged that:

- The user account belongs to the **VP of Global Finance**.
- The host machine targeted holds **PCI-compliant financial data**.



How SIEM Integrates the Ecosystem

Capabilities	Splunk Enterprise Security (SIEM)	EDR (Endpoint Detection)
Scope of Sight	Global Enterprise: Cloud, Network, Identity, Applications, Endpoints.	Localized Endpoint: Laptops, Workstations, and Virtual Machine servers.
Primary Mechanism	Cross-log data correlation and complex event aggregation.	Deep OS kernel behavioral inspection and memory tracking.
Data Types	Structured/Unstructured logs via Syslog, JSON, API, Event Channels.	Native raw telemetry (process creation, file modification streams).